

Mobile Coffee Cart Feasibility Study

**A Data-Driven Evaluation of Operating Strategy and
Demand Patterns**

**Report Submitted as Part of the IS 2010 Data Analytics
Challenge conducted by the University of Utah IS
program**

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Purpose

The purpose of this data analysis is to evaluate and understand the viability of a mobile coffee cart on the University of Utah campus. Determined via; optimal operating schedule, menu focus, resource planning, sales, weather, and foot traffic data.

Executive Summary

The University of Utah is considering launching a mobile coffee cart to serve students, faculty, and staff across campus. Before making a final investment decision, administrators need clarification on demand patterns, customer behavior, and operational requirements. Our analytics team conducted a comprehensive evaluation of simulated transaction data, weather conditions, and estimated campus foot traffic over the past year to determine when and where demand is strongest, which products drive revenue, and how external factors such as campus activity and temperature influence purchasing behavior. Our objective was to generate actionable, data backed recommendations that support the university's launch strategy and operational planning.

Our analysis confirms a strong business case for the coffee cart. Peak demand consistently occurs in the morning (7:00 am to 10:59 am) and early afternoon (11:00 am to 1:59 pm) time blocks, corresponding with high foot traffic during class transitions, as outlined in the time of day figures. During colder months, there is a greater demand for hot beverages, while iced drinks are significantly preferred in warmer seasons, demonstrating seasonal purchasing preferences. Additionally, sales positively correlate with estimated traffic, showing that campus activity levels reliably predict purchasing behavior. Weather further influences consumption, reinforcing the importance of seasonally adaptive menu planning and inventory management.

From these insights, we recommend operating the cart primarily during morning and midday hours, with potential reduction during summer months or low traffic periods, optimizing labor usage and reducing waste. Hot beverages should be highlighted during the fall and winter, and the marketing of iced beverages should be increased during warmer months. Using a random sample of transactions, we also estimated a confidence interval for the average daily sales, providing administrators with a reliable revenue range and a clear understanding of expected financial performance. Finally, through what if analysis, we identified the most profitable four hour operating window, aiding the university in strategically selecting a limited hour model. In conclusion, the data strongly support launching the coffee cart with a focused operating schedule and seasonally responsive menu strategy. With demand patterns that align with the predictable campus activity, the coffee cart has the potential to become profitable and add value for the University of Utah community.

Methodology

When determining whether a mobile coffee cart should operate on the University of Utah campus, our group first analyzed two datasets that were provided, this included: historic sales and weather along with estimated campus foot traffic.

Our first step was cleaning and preparing the data. Dates and times were standardized, and sales records were sorted and reviewed for accuracy. We created a time of day category (Morning, Midday, Afternoon, Evening) by grouping each transaction based on the hour it occurred. This allowed for clear comparisons across daily and weekly patterns. PivotTables were used to summarize total revenue and transaction volume by time of day, day of week, and season. Drinks were also grouped into hot and cold categories to observe shifts based on temperature. Daily revenue totals were calculated to measure overall performance and variation across the year.

To understand the relationship between foot traffic and sales, the foot traffic dataset was merged with aggregated sales using matching dates and time slots. We then calculated the correlation between traffic and transaction volume to determine how much campus activity drives sales. Finally, we calculated the average daily revenue and created a 95% confidence interval to estimate the expected daily performance of the cart under normal conditions.

Findings

Sales were highest during weekday mornings and mid day hours. The period from 7:00 a.m. to 1:59 p.m. accounted for the majority of annual revenue. Evenings and weekends showed low demand, aligning with typical campus traffic.

The most popular drinks were lattes, iced coffee, cold brew, and Americanos. Hot drinks made up the most purchases, and cold beverages became more popular during the summer. Overall, espresso remained strong across all seasons.

Weather had a limited impact on total sales. Warmer temperatures led to more cold drink purchases which is expected, but daily revenue did not change with temperature.

Foot traffic was a strong predictor of sales. The correlation between estimated foot traffic and transactions was approximately 79%, showing that the higher the foot traffic the more sales you'd expect to see.

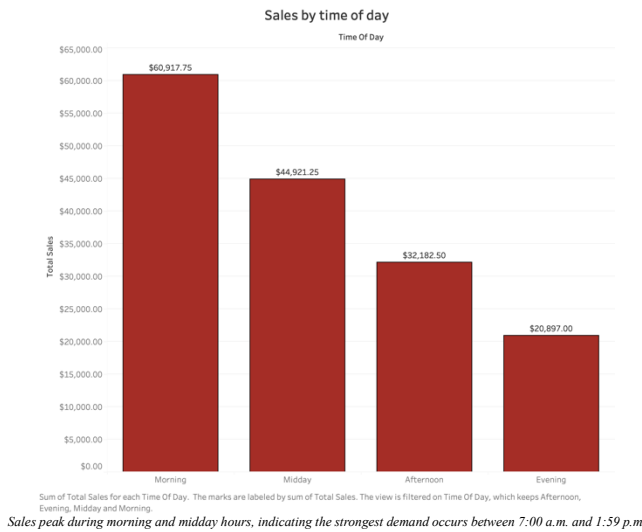
On average, the daily revenue was about \$448, with a range consisting of \$430 to \$470 based on the confidence interval. This suggests stable earnings for the coffee cart and also be used to supports staffing and inventory planning.

Recommendations

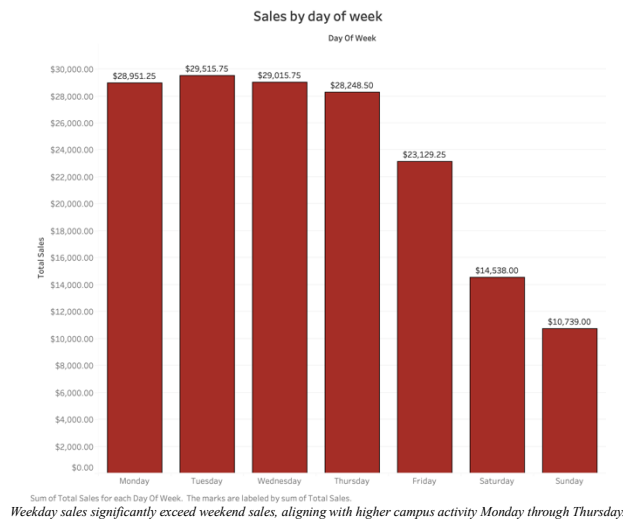
1. Operate Monday through Thursday from 7:00 a.m. to 2:00 p.m., when demand is strongest on campus.
2. Shorten Friday hours, as both traffic and sales decline after late morning due to many college students trying to avoid Friday classes.
3. Avoid weekend operations, as demand is consistently low.
4. If the coffee cart is limited to four hours per day, operate during 7:00–11:00 a.m., the most profitable time window.
5. Maintain a espresso based menu all year around
6. Increase the promotion of cold beverages during warm months (summer/ spring).
7. Use foot traffic patterns to guide staffing levels and inventory decisions to maximize the most profit.

Figures

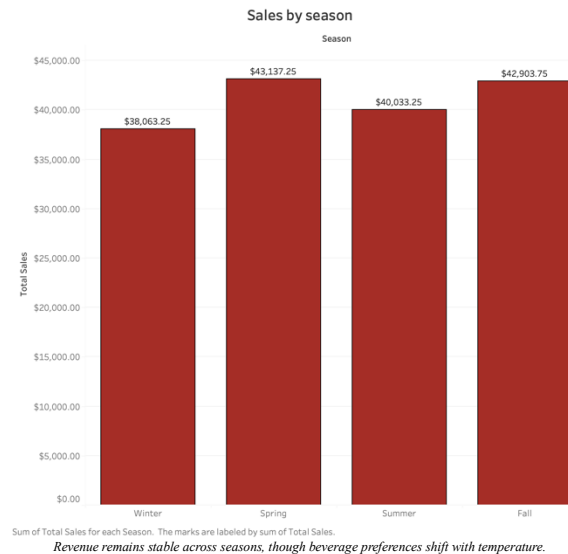
- Sales were highest during weekday mornings and midday hours, with the period from 7:00 a.m. to 1:59 p.m. producing the majority of annual revenue. These peaks align with heavy campus foot traffic during class transitions, indicating that operating during these hours would capture the strongest natural demand.



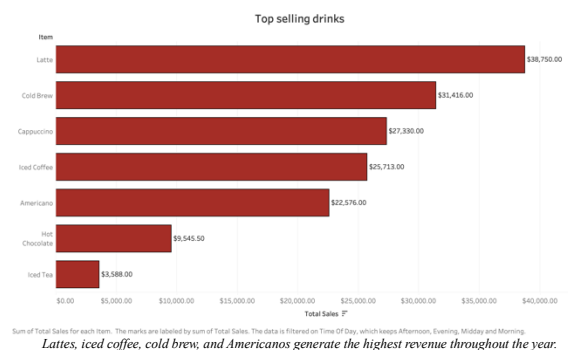
- Demand varied noticeably across the week, with Monday through Thursday generating the most consistent and reliable sales. Fridays showed a decline, and weekends demonstrated very low activity. This pattern reflects reduced campus presence outside the traditional academic week.



- Although beverage preferences shifted with temperature, total revenue remained stable across seasons. Hot drinks dominated in fall and winter, while iced beverages increased in spring and summer. These patterns reinforce the value of seasonal menu adjustments without requiring major operational changes.



- The most popular drinks were lattes, iced coffee, cold brew, and Americanos, all of which performed strongly across the year. Espresso based beverages consistently drove the largest share of revenue, highlighting the importance of maintaining a focused, core menu that emphasizes these high demand items.



- Weather had limited influence on overall daily revenue, but it did affect the mix of beverages sold. Warmer temperatures led to higher sales of iced drinks, while cooler weather increased demand for hot beverages. This shift supports strategic inventory planning based on seasonal conditions.

